

Optimization Techniques in Power System (BT EE610)



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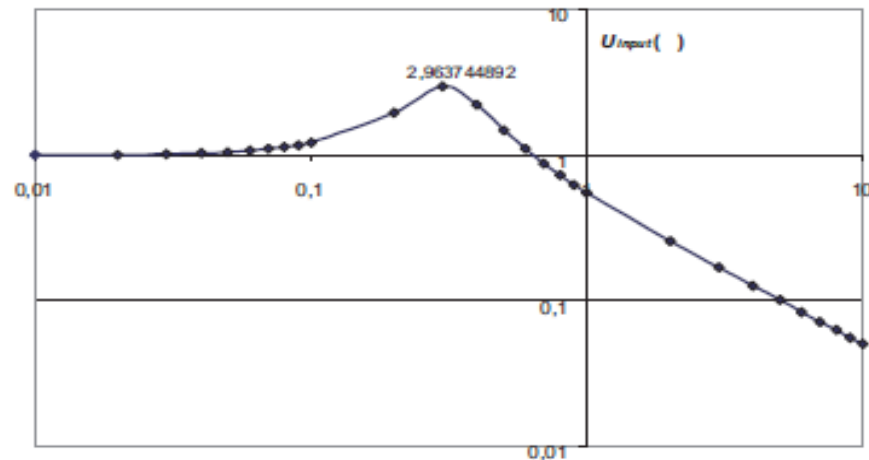
Unit-II Optimization in Electrical Engineering

- ❖ Introductory example
- ❖ Optimization environment
- ❖ Mathematical background
- ❖ Programming methods to solve optimization problems
- ❖ Power generation scheduling by integer programming

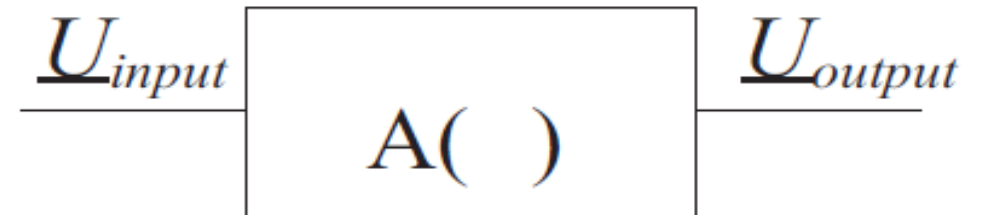
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1. INTRODUCTORY EXAMPLE

- Assume the amplitude of a voltage U_{input} as given in Fig. 1.1 (a) using the normalized frequency Ω . The required output voltage U_{output} should have a constant amplitude of 1V from very low frequencies to $\Omega = 1$. This can be done using an active filter as shown in Fig. 1.1 (b).



(a) Amplitude of the input voltage U_{input}



(b) Active filter

Figure 1.1: Active filter example

- A general active filter can be described mathematically by a complex function $A(P)$ of second order.

$$A(P) = \frac{d_0 + d_1(P) + d_2(P^2)}{c_0 + c_1(P) + c_2(P^2)}$$

- P can be set to be $j\Omega = j(f/f_c)$ with f_c being the *cut off* frequency.

$$A(j\Omega) = \frac{d_0 + d_1(P) - d_2(\Omega^2)}{c_0 + c_1(j\Omega) - c_2(\Omega^2)}$$

- The optimization/identification problem is to determine the coefficients $d_0, d_1, d_2, c_0, c_1, c_2$ of a filter, which fulfills the required operation.

2. OPTIMIZATION ENVIRONMENT

To solve one or all of the above mentioned problems, an optimization environment fulfilling the following requirements is needed.

- Analysis software to solve the forward problem.
- Optimization strategy that supplies the analysis software with new values for the optimization parameters and converge stable and fast towards a solution.
- Approximation algorithms to avoid the solution of the forward problem.
- Niching methods to detect "good" solutions besides the optimal one.
- Preprocessor checking the new values for the optimization parameters, transforming them in a suitable way.
- Possibility to assess the quality of the successively calculated forward problems via an objective function.
- Post-processor to visualize the obtained results.

3. MATHEMATICAL BACKGROUND

Optimization problems usually involve the minimization of an objective function $f(\mathbf{x})$, where $\mathbf{x}$ is a point in the optimization parameter space. The general problem class to be considered is known as a nonlinearly constraint optimization problem (NCP) and can be expressed in mathematical terms as :

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \tag{3.1}$$

$$\text{subject to } \hat{c}_i(\mathbf{x}) = 0 \quad i = 1, 2, \dots, m' - 1 \tag{3.2}$$

$$\hat{c}_i(\mathbf{x}) \geq 0 \quad i = m', \dots, m \tag{3.3}$$

The objective function $f(\mathbf{x})$ and (equality or inequality) constraint functions $c_i(\mathbf{x})$ are real value scalar functions. Any point $\hat{\mathbf{x}}$ satisfying all constraints of the optimization problem is called feasible. All feasible points are called the feasible region. If $N(\mathbf{x}^*, \delta)$ represents a set of feasible points in the neighborhood of a strong local minimizer $\mathbf{x}^*$, then (3.4) must hold:

$$f(\mathbf{x}^*) < f(\mathbf{x}) \quad \text{for all } \mathbf{x} \in N(\mathbf{x}^*, \delta) \quad \text{and} \quad \mathbf{x} \neq \mathbf{x}^*. \quad (3.4)$$

- **Local minima - global minimum:**

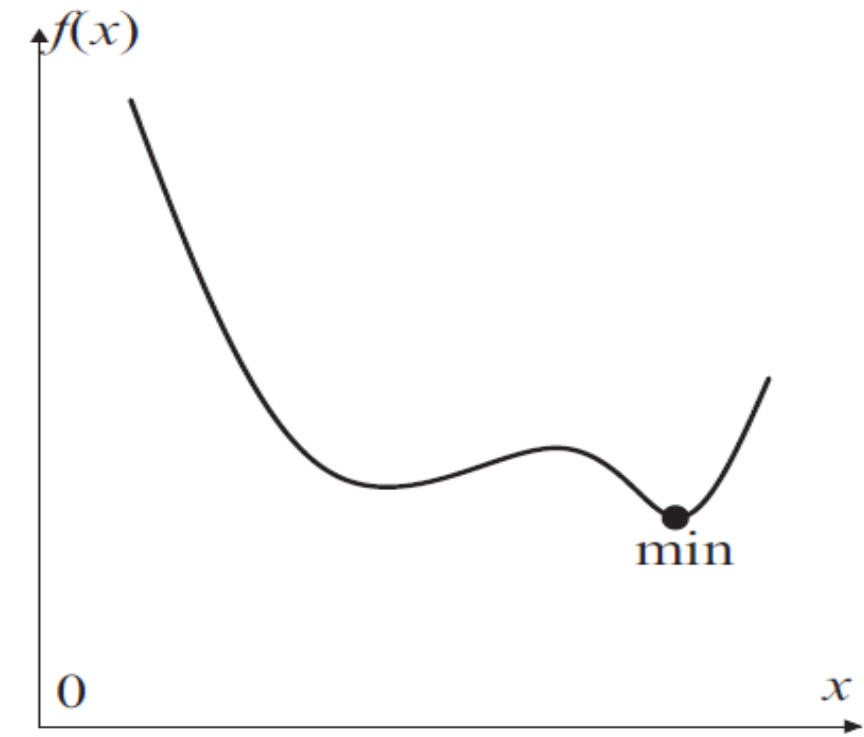
In real world application the feasible region very often contains one or more local minima besides the global minimum all of them fulfilling (3.4). In many problems it will be sufficient to discover a local minimum, nevertheless there exist applications where the global minimum is the only acceptable solution (identification problem).

- **Pareto-optimality in multi-objective optimization:**

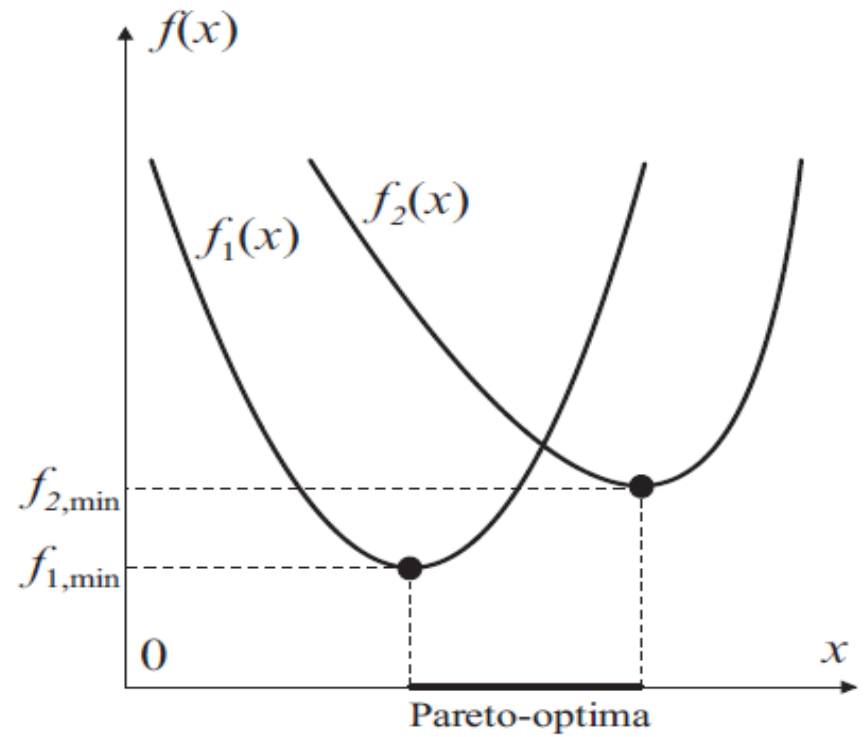
Most of the real-world optimization problems involve multiple conflicting objectives that must be mutually reconciled. These problems are called multi objective optimization problems (MOO) or vector optimization problems in contrast to single objective optimization (SOO) or scalar optimization problems.

A major characteristic of them is that in general the individual solutions for each single objective differ from each other and therefore no solution exists, where all solutions reach their individual minimum. The objective function $\mathbf{f}(\mathbf{x})$ of a vector optimization problem can be written as follows:

$$\min_{\mathbf{x} \in \mathbb{R}^n} \mathbf{f}(\mathbf{x}) = \min_{\mathbf{x} \in \mathbb{R}^n} (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x})) \quad (3.5)$$



(a) Single objective function $f(\mathbf{x})$



(b) Multi-objective function $f_1(\mathbf{x}), f_2(\mathbf{x})$

Fig. 3.1 optimal solutions

Fig. 3.1 shows both a one dimensional SOO and MOO problem. In general, there exists a set of pareto-optimal solutions rather than one unique solution for MOO problems.

4. PROGRAMMING METHODS TO SOLVE OPTIMIZATION PROBLEMS

- Linear Programming
- Quadratic Programming
- Least Square method
- Non-Linear
 - I. Constraints
 - II. Un-constraints
 - III. Equation Solving
 - IV. Curve Fitting

4.1 Linear Programming:

The optimization problem based on the nature of constraints functions and objective functions. If all functions g and u_i , ($i = 1, 2, \dots, m$) are linear functions of the decision variables y , then it is called Linear Optimization Problem or Linear Programming Problem (LPP). LPP can be solved by Graphical method, Algebra method, Simplex method, Big-m method, Two Phase method and so on.

$$\max \quad y_1 - 2y_2$$

subject to

$$y_1 + y_2 \leq 4$$

$$-2y_1 + y_2 \leq 3$$

$$y_1, y_2 \geq 0.$$

4.2 Non-linear Programming:

If either the objective function g or at least one of the constraints is non linear function of the decision variables y , then it is called nonlinear optimization problem or non-linear programming problem. For Example:

$$\max \quad z = 25y_1 + 30y_2 - 0.30y_2^2$$

subject to

$$y_1 + 0.2y_1^2 + 3y_2 + 0.35y_2^2 \leq 1200$$

$$y_1 + y_2 \leq 6000$$

$$y_1, y_2 \geq 0.$$

4.3 Integer Programming:

- An integer programming problem is a mathematical optimization or feasibility program in which all of the variables are restricted to be integers.
- Use of Integer program(gomory method) in real life. How the economic scheduling problem may be formulated as an integer linear program.

- It is one of the linear programming.
- Canonical Form of Integer Programming:

An integer linear program in canonical form is expressed as:

$$\max \quad c^T x$$

$$\text{Subjected} \quad Ax \leq b$$

$$x \geq 0$$

and $x \in \mathbb{Z}^n$

and an ILP in standard form is expressed as

$$\max \quad c^T x$$

subject to

$$Ax + s = b,$$

$$s \geq 0,$$

and $x \in \mathbb{Z}$

where c, b are vectors and A is a matrix, where all entries are integers.

4.3.1 Use of Integer Programming:

Now use this integer objective programming in power system problem.

Power system problem is of type :

- (1) Unit Commitment problem:** In unit power system; we can determine that according to requirement of electricity how much units should be switched on and off.
- (2) Economic Load Dispatch :** In a power system ; we can able to determine that how much allocation of generation must be given to different units for economic on.

5. POWER GENERATION SCHEDULING BY INTEGER PROGRAMMING

- In this, the economic scheduling of electric generator is taken. It include the selection of unit to operate at any time and power as per requirement to each unit.

This economic dispatch problem includes the following :

1. The discontinuous power output characteristics of the generators, that is it has either no power output, or a minimum output.
2. The cost of on and off of each unit.
3. The dispatch of load by incremental costs.

Steam-turbine generating units include minimum three types of non-linearities in the Rs./dollar per- hour input to power output characteristics.

1. First one is cost incurred when the turbine is started;
2. Second one is the output constrained to lie between the highest and lowest ratings; and
3. Last is input-output relation complicated by the presence of valve loops.

So to formulate this, an example is used.

- Suppose a power system with two generators. The startup costs, minimum outputs costs, and incremental costs are given in the following table:

Table 5.1 Generator cost characteristics

Symbol	Name	Generator 1	Generator 2
U_j	Maximum Rating	80 mw	120 mw
L_j	Minimum Rating	20 mw	30 mw
$c_j L$	Cost at minimum rating	\$60/hr	\$85/hr
$c_j l$	1st incremental cost step	\$2.3/mw-hr	\$2.0/mw-hr
Δi_l	Power available at c_{i_j}	30mw	20mw
c_{i_2}	2nd incremental Step	\$3.0/mw-hr	\$2.8/mw-hr
Δi_2	Power available at c_{i_2}	30mw	70mw
c_{i_u}	Start-up cost	\$20	\$30
c_{i_d}	Shutdown cost	\$10	\$15

Table 5.2 Capacity requirement

P_t	Estimated load	50mw	100mw
	Spinning reserve	15mw	20mw
S_t	Total Requirement	65mw	120mw

Such characteristics has been found to given a lower cost schedule than connected straight line segments when the effects of valve loops are considered.

a) The on-off indicator for generator i in time period will be:

$y_{jt} = 1$ if generator is scheduled to operate

$y_{jt} = 0$ otherwise

b) The start up indicator for generator i in period t will be:

$w_{jt} = 1$ if generator is scheduled to start

$w_{jt} = 0$ otherwise

c) The shut down indicator for generator i in period t will be:

$v_{jt} = 1$ if generator is scheduled to start

$v_{jt} = 0$ otherwise

d) The output from generator i at incremental cost step j in period t will be :

$x_{jkt} = mw$

The solution must satisfy the following requirements:

1. All nonnegative variables are present.
2. Assume all variables have integer values.
3. The total capacity required for each period must have less than or equal to operating units capacity,

$$80y_{11} + 120y_{21} \geq 65$$

$$80y_{12} + 120y_{21} \geq 120$$

4. From each unit, the sum of the power output must be equal to estimated load, D_t .

$$20y_{11} + x_{111} + x_{121} + 30y_{21} + x_{211} + x_{221} = 50$$

$$20y_{12} + x_{112} + x_{122} + 30y_{22} + x_{212} + x_{222} = 100$$

5. At each and every incremental cost step, the power output from each generator must be less than, or equal to, the maximum available capacity, and if the generator is not operating then it must be zero.

$$30y_{11} \geq x_{111}, 30y_{12} \geq x_{112}$$

$$30y_{11} \geq x_{121}, 30y_{12} \geq x_{122}$$

$$20y_{21} \geq x_{211}, 20y_{22} \geq x_{212}$$

$$70y_{21} \geq x_{221}, 70y_{22} \geq x_{222}$$

6. The start indicator for that period, v_{it} must be 1, when generator started. The shutdown indicator for that period, w_{it} must be 1 when generator is off (shutdown).

$$y_{11} - y_{10} = w_{11} - v_{11}$$

$$y_{21} - y_{20} = w_{21} - v_{21}$$

$$y_{12} - y_{11} = w_{12} - v_{12}$$

$$y_{22} - y_{20} = w_{22} - v_{22}$$

7. For each generator, the on-off indicator must be not greater than 1, in each period.

$$y_{11} \leq 1, y_{21} \leq 1$$

$$y_{12} \leq 1, y_{22} \leq 1$$

8. The object is to minimize the cost function for each time period, where the cost function is the sum of the costs of stopping, starting, producing power at minimum output, and producing power above the minimum output. Let the cost be represented by the symbol z .

$$\begin{aligned}
z = & 20w_{11} + 30w_{21} + 10v_{11} + 15v_{21} + 60y_{11} + 85y_{12} + 2.3x_{111} + 3.0x_{121} \\
& + 2.0x_{211} + 2.8x_{221} + 20w_{12} + 30w_{22} + 10v_{12} + 15v_{22} + 60y_{12} + 85y_{22} \\
& + 2.3x_{112} + 3.0x_{122} + 2.0x_{212} + 2.8x_{222}
\end{aligned}$$

This completes the formulation of above problem. The generator scheduling example does not have start up costs which vary with length of the time generator has been shut down. Additional equations should be also noticed, specially generator requirements, such as keeping one unit at the station on one line, could be entered as further constraint without affecting the method of solution.

5.1 Method of Solution:

- Firstly, the economic scheduling of generators is mathematically solved by assuming that all variables have integer values. The problem, without the integer requirement is a linear programming problem solvable by dual method or simplex method.
- Then in 1958 to solve linear program with additional integer requirements was presented by R. E. Gomory. The derivation of additional constraint from already present in the problem is the key to his method. This addition of the constraint will allow all integer solutions to the original problem, but eliminate the non integer solutions.

- To solve generator scheduling problem gomory hand calculations led to discovery of a more general algorithm and original integer algorithm which is named eucledian escalator method. Referred to the dual eucledian method and is identical to the method presented by gomory in 1960. So first write this formulated equations in standard form.

In standard form:

$$80y_{11} + 120y_{21} - S_1 = 65$$

$$80y_{12} + 120y_{22} - S_2 = 120$$

$$30y_{11} - x_{111} - S_3 = 0$$

$$30y_{12} - x_{112} - S_4 = 0$$

$$30y_{11} - x_{121} - S_5 = 0$$

$$30y_{12} - x_{122} - S_6 = 0$$

$$20y_{21} - x_{211} - S_7 = 0$$

$$20y_{22} - x_{212} - S_8 = 0$$

$$70y_{21} - x_{221} - S_9 = 0$$

$$70y_{22} - x_{222} - S_{10} = 0$$

$$0 = y_{11} - y_{10} - w_{11} + v_{11}$$

$$0 = y_{21} - y_{20} - w_{21} + v_{21}$$

$$0 = y_{12} - y_{11} - w_{21} + v_{12}$$

$$0 = y_{22} - y_{21} - w_{22} + v_{22}$$

$$y_{11} + S_{11} = 1$$

$$y_{21} + S_{12} = 1$$

$$y_{12} + S_{13} = 1$$

$$y_{22} + S_{14} = 1$$

$$20y_{11} + x_{111} + x_{121} + 30y_{21} + x_{211} + x_{221} + S_{15} = 50$$

$$20y_{12} + x_{112} + x_{122} + 30y_{22} + x_{212} + x_{222} + S_{16} = 100$$

Table 5.4 Final table

[illegible]

Thank You